

Practical 3

MaxHeap Sort

Theory :-

Heap Sort is a comparison-based sorting algorithm that uses a Binary Heap data structure. For sorting in ascending order, a Max Heap is constructed, where every parent node is greater than or equal to its child nodes.

Code :-

```
#include <iostream>
using namespace std;
#define MAX_SIZE 100

void Heapify(int arr[], int n, int i){
    int l = 2*i + 1;
    int r = 2*i + 2;
    int largest = i;

    if(l < n && arr[l] > arr[largest]){
        largest = l;
    }

    if(r < n && arr[r] > arr[largest]){
        largest = r;
    }

    if(largest != i){
        swap(arr[i], arr[largest]);
        Heapify(arr, n, largest);
    }
}

void maxHeapSort(int arr[], int n){
    for(int i = n/2 - 1; i >= 0; i--){
        Heapify(arr, n, i);
    }

    for(int i = n - 1; i > 0; i--){
        swap(arr[0], arr[i]);
        Heapify(arr, i, 0);
    }
}
```

```

void printArray(int arr[], int n ){
    for (int i = 0 ; i < n ; i++){
        cout << arr[i] << " ";
    }
}

int main(){
    int n ;
    cout << "Enter the size of array : ";
    cin >> n ;

    int arr[MAX_SIZE];

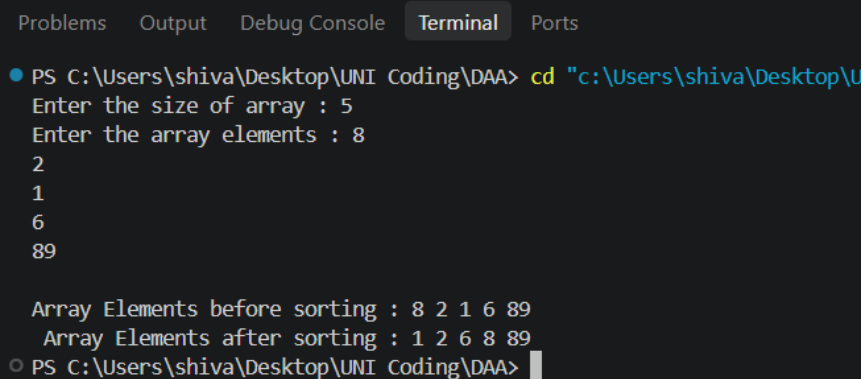
    // Taking input from user
    cout << "Enter the array elements : ";
    for (int i = 0 ; i < n ; i++){
        cin >> arr[i];
    }
    cout << endl ;

    // printing array elements before sorting
    cout << "Array Elements before sorting : ";
    printArray(arr,n);
    maxHeapSort(arr,n);
    cout << "\n Array Elements after sorting : ";
    printArray(arr,n);

    return 0;
}

```

Output :-



```

Problems  Output  Debug Console  Terminal  Ports
● PS C:\Users\shiva\Desktop\UNI Coding\DAA> cd "c:\Users\shiva\Desktop\U
Enter the size of array : 5
Enter the array elements : 8
2
1
6
89

Array Elements before sorting : 8 2 1 6 89
Array Elements after sorting : 1 2 6 8 89
○ PS C:\Users\shiva\Desktop\UNI Coding\DAA>

```

Analysis Of Algorithm :- MaxHeap Sort

- **Best Case:** $O(n \log n)$
- **Average Case:** $O(n \log n)$
- **Worst Case:** $O(n \log n)$
- **Space Complexity:** $O(\log n)$ due to recursive `Heapify()`.
- **Auxiliary Space:** $O(1)$ excluding the recursive call stack.
- **Stability:** Not stable.
- **Sorting Type:** In-place sorting algorithm.
- **Overall Complexity:**